

40.520 Stochastic Models

Continuous Time Markov Chain

Xiaotang Yang

Engineering Systems and Design (ESD)
Singapore University of Technology and Design (SUTD)

Spring 2026

Outline

1. Introducing Continuous -Time Markov Chain
2. Kolmogorov Backward/Forward Equations
3. Limiting Probabilities
 - Embedded Markov Chain
 - Uniformization
4. Classical Queueing Examples.

Continuous-Time Markov Chain

Recap: Discrete-Time Markov Chains (DTMC)

Definition (DTMC)

A time-homogeneous DTMC (Discrete-Time Markov Chain) is a stochastic process $\{X_n, n = 0, 1, 2, \dots\}$, where X_n denotes the state at (discrete) time step n and such that, $\forall n \geq 0$, $\forall i, j$, and $\forall i_0, \dots, i_{n-1}$:

$$\begin{aligned} & \mathbf{P}\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0\} \\ &= \mathbf{P}\{X_{n+1} = j \mid X_n = i\} \\ &= P_{ij}. \end{aligned}$$

Three Properties of A DTMC

1. Transitions are made at discrete time steps, $n = 0, 1, 2, \dots$
2. The past does not matter. Only the present state matters.
3. Transition probabilities are “stationary”.

Continuous-Time Markov Chain (CTMC)

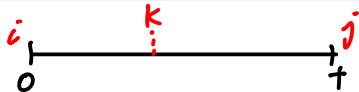
Definition (CTMC)

A time-homogeneous Continuous-Time Markov Chain (CTMC) is a continuous-time stochastic process $\{X(t), t \geq 0\}$ such that, $\forall s, t \geq 0$ and $\forall i, j, x(u)$:

$$\begin{aligned} & \mathbf{P}\{X(t+s) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s\} \\ &= \mathbf{P}\{X(t+s) = j \mid X(s) = i\} \\ &= \mathbf{P}\{X(t) = j \mid X(0) = i\} = P_{ij}(t) \end{aligned}$$

Three Properties of A ~~CTMC~~ CTMC

- Transitions between states can happen at any time.
- Markovian Property: Only the present state matters.
- Transition probabilities are “stationary”.



Memoryless Holding Times

Holding Time at state i . Define τ_i to be the time until the CTMC leaves state i , given that the CTMC is currently in state i . Then

$$\mathbf{P}\{\tau_i > t + s \mid \tau_i > s\} = \mathbf{P}\{\tau_i > t\}$$

Implication

- τ_i is memoryless
- τ_i follows exponential distribution

$$P(\tau_i > t+s \mid \tau_i > s) = P(X_u = i, \forall u \in [0, t+s] \mid X_u = i, \forall u \in [0, s])$$

$$= P(X_u = i, \forall u \in [s, t+s] \mid X_u = i, \forall u \in [0, s])$$

$$= P(X_u = i, \forall u \in [s, t+s] \mid X_s = i)$$

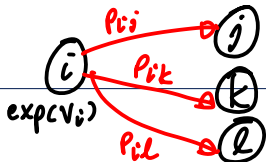
$$= P(X_u = i, \forall u \in [0, t] \mid X_0 = i)$$

$$= P(\tau_i > t)$$

Markov's
property

stationary

CTMC - View 1



Definition (CTMC - View 1)

A CTMC is a stochastic process with the property that every time it enters state i , the following hold:

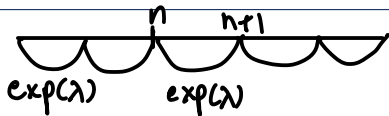
1. The amount of time the process spends in state i before making a transition is Exponentially distributed with rate ν_i .
2. When the process leaves state i , it will next enter state j with **jump probability** p_{ij} independent of the time spent at state i .

Remark: We have a Discrete-Time Markov Chain embedded in the process with transition probabilities being the jump probabilities p_{ij} .

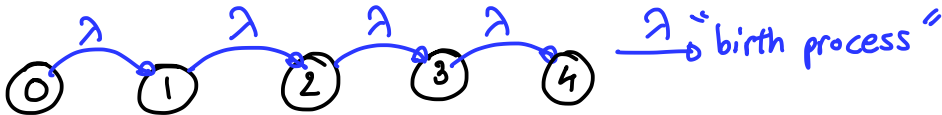
Embedded Markov Chain

The **embedded Markov chain** is the discrete-time Markov chain formed by the sequence of states visited by a continuous-time Markov chain (CTMC) *at the moments of transition*.

A New Perspective on Poisson Process



- A Poisson process with rate λ can be viewed as a CTMC:
 - At each state n , it will stay for an exponential time with mean $1/\lambda$;
 - Then, it proceeds from n to $n+1$.



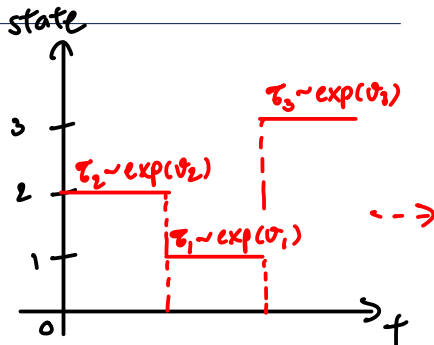
An Example

- $S = \{1, 2, 3\}$
- “Jump” matrix

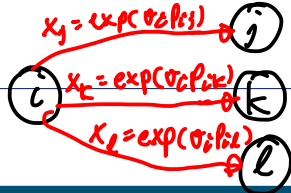
$$\begin{pmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}$$

- Holding time rates

$$v_1 = 3, \quad v_2 = 2, \quad v_3 = 1$$



CTMC - View 2



Definition (CTMC - View 2)

Let $X_j \sim \text{Exp}(\nu_i p_{ij})$ represent the time to transition from i to j , $\forall j \neq i$. Let $\tau_i = \min_j \{X_j\}$ be the time until the Markov chain leaves state i . Let the next state be m where $m = \operatorname{argmin}_j \{X_j\}$, i.e., X_m was the $\min_j \{X_j\}$.

Interpretation

- Being in state i and waiting for multiple independent “calls” from other states
- The time until a call from state j is $X_j \sim \text{Exp}(\nu_i p_{ij})$.
- The first call received determines the next state.

Q-Matrix

The **rate matrix** or **infinitesimal generator** $\mathbf{Q} = [Q_{ij}]$ describes the transition rates of a continuous-time Markov chain.

- $Q_{ij} = q_{ij} = p_{ij}\nu_i$ for $i \neq j$
- $Q_{ii} = -\nu_i = -\sum_{j \neq i} q_{ij}$. The diagonal entries are negative and represent the *total exit rate* from state i .

$$\mathbf{Q} = \begin{pmatrix} -\nu_1 & p_{12}\nu_1 & p_{13}\nu_1 & \cdots \\ p_{21}\nu_2 & -\nu_2 & p_{23}\nu_2 & \cdots \\ p_{31}\nu_3 & p_{32}\nu_3 & -\nu_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where $\sum_{j \neq i} p_{ij} = 1$.

Q-Matrix

- The **Q** matrix fully characterizes the CTMC: rates determine both holding times and jump probabilities.
- Relationship: **Q**-Matrix and transition matrix **P** of the embedded Markov chain:

$$p_{ij} = \frac{Q_{ij}}{\nu_i} \quad (i \neq j)$$

↓
for off-diagonal entries

↓
For diagonal entries: $p_{ii} = 0 \neq$

Example

$$p_{ij} = \frac{Q_{ij}}{\sigma_i}, \quad i \neq j$$

$$\sigma_1 = \lambda_{12} + \lambda_{13}$$

$$\sigma_2 = \lambda_{21} + \lambda_{23}$$

$$\sigma_3 = \lambda_{31} + \lambda_{32}$$

$$p_{ii} = 0$$

Given the Q-Matrix of a CTMC:

$$Q = \begin{pmatrix} -(\lambda_{12} + \lambda_{13}) & \lambda_{12} & \lambda_{13} \\ \lambda_{21} & -(\lambda_{21} + \lambda_{23}) & \lambda_{23} \\ \lambda_{31} & \lambda_{32} & -(\lambda_{31} + \lambda_{32}) \end{pmatrix}$$

find the transition matrix of the embedded Markov chain.

$$\therefore P = \begin{pmatrix} 0 & \frac{\lambda_{12}}{\lambda_{12} + \lambda_{13}} & \frac{\lambda_{13}}{\lambda_{12} + \lambda_{13}} \\ \frac{\lambda_{21}}{\lambda_{21} + \lambda_{23}} & 0 & \frac{\lambda_{23}}{\lambda_{21} + \lambda_{23}} \\ \frac{\lambda_{31}}{\lambda_{31} + \lambda_{32}} & \frac{\lambda_{32}}{\lambda_{31} + \lambda_{32}} & 0 \end{pmatrix}$$

Kolmogorov Backward/Forward Equations

Transition Probability Function

$p_{ij}(t)$ w.o. $t \rightarrow \infty \neq \pi_j$

- Let $\{X_t, t \geq 0\}$ record the state of a CTMC over time. Then

$$P_{ij}(t) = P(X_{t+s} = j \mid X_s = i)$$

is known as the transition probability function of the process, i.e., the probability of being in state j at time t , given that the process started in state i at time 0.

- We have

$$P_{ij}(t+s) = \sum_{k=1}^n P_{ik}(t)P_{kj}(s).$$

$$\begin{aligned}
 p_{ij}(t+s) &= P(X_{t+s}=j \mid X_0=i) \\
 &= \sum_{k=1}^n P(X_{t+s}=j, X_t=k \mid X_0=i) \\
 &= \sum_{k=1}^n \frac{P(X_{t+s}=j, X_t=k, X_0=i)}{P(X_0=i)} \\
 &= \sum_{k=1}^n \frac{P(X_{t+s}=j, X_t=k, X_0=i)}{P(X_t=k, X_0=i)} \cdot \frac{P(X_t=k, X_0=i)}{P(X_0=i)}
 \end{aligned}$$

Markov's property

$$\begin{aligned}
 &= \sum_{k=1}^n P(X_{t+s}=j \mid X_t=k, X_0=i) P(X_t=k \mid X_0=i) \\
 &= \sum_{k=1}^n P(X_{t+s}=j \mid X_t=k) \cdot P(X_t=k \mid X_0=i)
 \end{aligned}$$

stationary

$$= \sum_{k=1}^n P(X_s=j \mid X_0=k) \cdot P(X_t=k \mid X_0=i) = \sum_{k=1}^n p_{kj}(s) p_{ik}(t) \quad \neq$$

Kolmogorov Backward/Forward Equations

Theorem (Kolmogorov Backward Equation) For states i, j , and times $t \geq 0$,

$$P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - \nu_i P_{ij}(t),$$

where

$$q_{ik} = \nu_i p_{ik}.$$

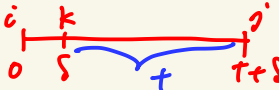
$$P'_{ij}(t) = \lim_{\delta \rightarrow 0} \frac{P_{ij}(t+\delta) - P_{ij}(t)}{\delta}$$

Backward:

$$\begin{aligned} P'_{ij}(t) &= \lim_{\delta \rightarrow 0} \frac{P_{ij}(t+\delta) - P_{ij}(t)}{\delta} \\ &= \lim_{\delta \rightarrow 0} \frac{\sum_{k=1}^n P_{ik}(\delta) P_{kj}(t) - P_{ij}(t)}{\delta} \end{aligned}$$

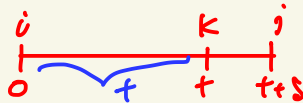
$$= \lim_{\delta \rightarrow 0} \frac{\sum_{k \neq i} P_{ik}(\delta) P_{kj}(t) + P_{ii}(\delta) \cdot P_{ij}(t) - P_{ij}(t)}{\delta}$$

Backward:



$$P_{ij}(t+\delta) = \sum_{k=1}^n P_{ik}(\delta) P_{kj}(t)$$

Forward:



$$P_{ij}(t+\delta) = \sum_{k=1}^n P_{ik}(t) P_{kj}(\delta)$$

$$\begin{aligned}
 &= \lim_{\delta \rightarrow 0} \frac{\sum_{k \neq i} P_{ik}(\delta) P_{kj}(t) - (1 - P_{ii}(\delta)) P_{ij}(t)}{\delta} \\
 &= \sum_{k \neq i} \left[\lim_{\delta \rightarrow 0} \frac{P_{ik}(\delta)}{\delta} P_{kj}(t) - \lim_{\delta \rightarrow 0} \left(\frac{1 - P_{ii}(\delta)}{\delta} \right) P_{ij}(t) \right] = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i P_{ij}(t) \quad \neq
 \end{aligned}$$

$P_{ik}(\delta) = P(\tau_i < \delta, m=k) + O(\delta^2)$ $\rightarrow$ Probability of having 2 more transaction with δ
 $= (1 - e^{-v_i \delta}) P_{ik} + O(\delta^2)$

$$\begin{aligned}
 &= [1 - (1 - v_i \delta)] P_{ik} + O(\delta) \\
 &= v_i \delta P_{ik} + O(\delta)
 \end{aligned}
 \quad \left\{ \begin{aligned} \lim_{\delta \rightarrow 0} \frac{P_{ik}(\delta)}{\delta} &= \lim_{\delta \rightarrow 0} \frac{(v_i P_{ik} \delta + O(\delta^2))}{\delta} = v_i P_{ik} = q_{ik} \end{aligned} \right.$$

$$\begin{aligned}
 P_{ii}(\delta) &= P(\tau_i > \delta) + O(\delta^2) \\
 &= e^{-v_i \delta} + O(\delta^2) \\
 &= 1 - v_i \delta + O(\delta)
 \end{aligned}
 \quad \left\{ \begin{aligned} \lim_{\delta \rightarrow 0} \left(\frac{1 - P_{ii}(\delta)}{\delta} \right) &= \lim_{\delta \rightarrow 0} \left(\frac{1 - (1 - v_i \delta + O(\delta^2))}{\delta} \right) = v_i \end{aligned} \right.$$

Kolmogorov Backward/Forward Equations

Theorem (Kolmogorov Forward Equation) For states i, j , and times $t \geq 0$,

$$P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - \nu_j P_{ij}(t).$$

Forward:

$$P_{ij}'(t) = \lim_{\delta \rightarrow 0} \frac{P_{ij}(t+\delta) - P_{ij}(t)}{\delta}$$

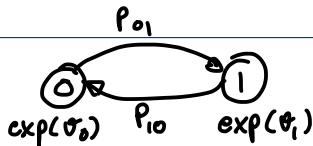
$$= \lim_{\delta \rightarrow 0} \frac{\sum_{k \neq i}^n P_{ik}(t) \cdot P_{kj}(\delta) - P_{ij}(t)}{\delta}$$

$$= \lim_{\delta \rightarrow 0} \frac{\sum_{k \neq j} P_{ik}(t) \cdot P_{kj}(\delta) - (1 - P_{jj}(\delta)) P_{ij}(t)}{\delta}$$

$$= \sum_{k \neq j} \left[\underbrace{\lim_{\delta \rightarrow 0} \frac{P_{kj}(\delta)}{\delta}}_{q_{kj}} \cdot P_{ik}(t) - \underbrace{\lim_{\delta \rightarrow 0} \left(\frac{1 - P_{jj}(\delta)}{\delta} \right)}_{\vartheta_j} P_{ij}(t) \right]$$

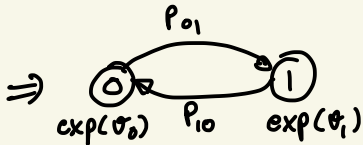
Example: Two-State Chain

state 0 = working
state 1 = breaking



- Consider a machine that works for an exponential amount of time having mean $1/\lambda$ before breaking down; and suppose that it takes an exponential amount of time having mean $1/\mu$ to repair the machine. $\theta_0 = \lambda, \theta_1 = \mu$
- If the machine is in working condition at time 0, then what is the probability that it will be working at time $t = 10$? $\rightarrow p_{00}(10) = ?$

state 0 = working
state 1 = breaking



$$\Rightarrow Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

Backward:

$$p'_{00}(t) = q_{01} p_{10}(t) - v_0 p_{00}(t) = \lambda p_{10}(t) - \lambda p_{00}(t) \quad \text{--- (1)}$$

$$p'_{10}(t) = q_{10} p_{00}(t) - \theta_1 p_{10}(t) = \mu p_{00}(t) - \mu p_{10}(t) \quad \text{--- (2)}$$

$$\mu \times \text{①} + \lambda \times \text{②} ; \mu p'_{00}(t) + \lambda p'_{10}(t) = 0 \Rightarrow [\mu p_{00}(t) + \lambda p_{10}(t)]' = 0$$

$$\therefore \mu p_{00}(t) + \lambda p_{10}(t) = \text{constant}$$

$$\because t=0 \rightarrow X_0=0 \rightarrow \therefore p_{00}(0)=1, p_{10}(0)=0 \Rightarrow \therefore \text{constant} = \mu$$

$$\therefore \mu p_{00}(t) + \lambda p_{10}(t) = \mu$$

$$\Rightarrow \boxed{p_{10}(t) = \frac{\mu - \mu p_{00}(t)}{\lambda}}$$

$$\therefore P_{00}'(t) = \lambda P_{10}(t) - \lambda P_{00}(t)$$

$$P_{00}'(t) = \lambda \left(\frac{\mu - \mu P_{00}(t)}{\lambda} \right) - \lambda P_{00}(t)$$

$$P_{00}'(t) = \mu - (\lambda + \mu) P_{00}(t) \rightarrow \text{Solve by 1}^{\text{st}} \text{ Order Non-homogeneous Linear ODE}$$

$$\therefore P_{00}'(t) + \underbrace{(\lambda + \mu)}_{p(t)} P_{00}(t) = \underbrace{\mu}_{f(t)} \Rightarrow \therefore P_{00}(t) = \frac{\mu}{\lambda + \mu} e^{(\lambda + \mu)t} e^{-(\lambda + \mu)t} + A e^{-(\lambda + \mu)t}$$

$$\therefore P(t) = -(\lambda + \mu)t$$

$$v'(t) = e^{-P(t)} f(t) = \mu e^{(\lambda + \mu)t}$$

$$\therefore v(t) = \frac{\mu}{\lambda + \mu} e^{(\lambda + \mu)t}$$

$$\therefore P_{00}(0) = 1 \Rightarrow \therefore A = \frac{\lambda}{\lambda + \mu}$$

$$\therefore P_{00}(t) = \frac{\mu}{\mu + \lambda} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t}$$

#

First Order Non-Homogeneous Linear ODE

Given a first order non-homogeneous linear differential equation

$$y' + p(t)y = f(t),$$

using variation of parameters the general solution is given by

$$y(t) = v(t)e^{P(t)} + Ae^{P(t)},$$

where $v'(t) = e^{-P(t)}f(t)$ and $P(t)$ is an antiderivative of $-p(t)$.

Proof: Linear ODE closed-form solution

From $y'(t) + p(t)y(t) = f(t)$, we multiply by $\mu(t)$ (Integration Factor) s.t.

$$(\mu(t)y(t))' = \mu(t)y'(t) + \mu'(t)y(t).$$

We get

$$\mu(t)y'(t) + \mu(t)p(t)y(t) = \mu(t)f(t);$$

therefore,

$$\mu'(t) = \mu(t)p(t),$$

$$\int \frac{d\mu(t)}{\mu(t)} = \int p(t) dt \triangleq -P(t),$$

$$\ln |\mu(t)| = -P(t),$$

$$\mu(t) = e^{-P(t)}.$$

Hence,

$$(u(t)y(t))' = e^{-p(t)}f(t),$$

$$u(t)y(t) = \int e^{-p(t)}f(t) dt + A \triangleq v(t) + A,$$

$$e^{-p(t)}y(t) = v(t) + A,$$

$$\therefore y(t) = v(t)e^{p(t)} + Ae^{p(t)},$$

#

General CTMC

- For any pair of states i and j , let

$$Q_{ij} = \begin{cases} q_{ij}, & i \neq j \\ -v_i, & i = j \end{cases}.$$

- Then, we can rewrite the backward/forward equations as follows:

$$P'_{ij}(t) = \sum_k Q_{ik} P_{kj}(t) \quad (\text{backward})$$

$$P'_{ij}(t) = \sum_k P_{ik}(t) Q_{kj} \quad (\text{forward})$$

Computing Transition Probabilities

- Introduce matrices Q , $P(t)$, $P'(t)$ by letting the element in row i , column j of these matrices be, respectively, Q_{ij} , $P_{ij}(t)$, $P'_{ij}(t)$.
- The two equations can be written as:

$$\begin{aligned}P'(t) &= QP(t) \quad (\text{backward}) \\P'(t) &= P(t)Q \quad (\text{forward})\end{aligned}$$

- The solution should be

$$P(t) = \sum_{n=0}^{+\infty} Q^n \frac{t^n}{n!}.$$

Limiting Probabilities

Limiting Probabilities

- The probability that a continuous-time Markov chain will be in state j at time t often converges to a limiting value that is independent of the initial state.
- We intend to study

$$\pi_j = \lim_{t \rightarrow +\infty} P_{ij}(t) = ?$$

DTMC

1. Irreducible
2. Positive recurrent
3. Aperiodic

CTMC

Limiting Probabilities

- To derive a set of equations for the π_i , we may let t approach ∞ in the forward equation. Then,

$$0 = \sum_{k \neq j} q_{kj} \pi_k - \nu_j \pi_j.$$

- In addition,

$$\sum_i \pi_i = 1.$$

Forward: $P_{ij}'(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - \vartheta_j P_{ij}(t)$

LHS: $\lim_{t \rightarrow \infty} P_{ij}'(t) = \lim_{t \rightarrow \infty} \frac{d}{dt} P_{ij}(t)$
 $= \frac{d}{dt} \lim_{t \rightarrow \infty} P_{ij}(t)$
 $= \frac{d}{dt} \pi_j = 0$

RHS: $\lim_{t \rightarrow \infty} \left[\sum_{k \neq j} q_{kj} P_{ik}(t) - \vartheta_j P_{ij}(t) \right] = \sum_{k \neq j} q_{kj} \pi_k - \vartheta_j \pi_j$

$$\therefore 0 = \sum_{k \neq j} q_{kj} \pi_k - \vartheta_j \pi_j$$

$$\vartheta_j \pi_j = \sum_{k \neq j} q_{kj} \pi_k$$

Limiting Probabilities

$\tilde{\pi}_i$ = the limiting distribution for the CTMC

- It is easy to see that if the embedded Markov chain has a limiting stationary distribution

$$\pi = (\pi_1, \dots, \pi_n), \quad \text{i.e.,} \quad \pi_i = \sum_j \pi_j p_{ji},$$

then

$$\tilde{\pi}_i \propto \frac{\pi_i}{v_i}.$$

- The solution to the equations on the last slide should be

$$\tilde{\pi}_i = \frac{\pi_i / v_i}{\sum_i \pi_i / v_i}.$$

Balance equation for the CTMC

$$\sum_{k \neq j} \tilde{\pi}_k q_{kj} = \tilde{\pi}_j \sigma_j$$

$$\sum_{k \neq j} \tilde{\pi}_k p_{kj} \vartheta_k = \tilde{\pi}_j \vartheta_j$$

compare these 2 equations

Embedded DTMC

$$\pi_j = \sum_{k \neq j} \pi_k p_{kj}$$

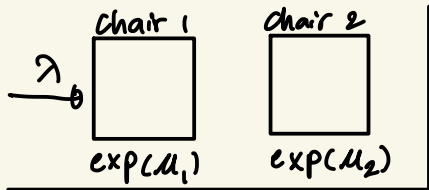
$$\tilde{\pi}_j \propto \frac{\pi_j}{\vartheta_j}, \exists c > 0 \text{ s.t. } \tilde{\pi}_j = c \frac{\pi_j}{\vartheta_j}$$

$$\sum \tilde{\pi}_j = \sum c \frac{\pi_j}{\vartheta_j} \Rightarrow \therefore c = \frac{1}{\sum \pi_j / \vartheta_j}$$

$$\therefore \tilde{\pi}_j = \frac{\pi_j / \vartheta_j}{\sum \pi_j / \vartheta_j}$$

Example: A Shoe Shine Shop

- Consider a shoe shine establishment consisting two chairs — chair 1 and chair 2. A customer upon arrival goes initially to chair 1 where his shoes are cleaned and polish is applied. After this is done, the customer moves on to chair 2 where the polish is buffed.
- The service times at the two chairs are independent and exponentially distributed with respective rates μ_1 and μ_2 .
- Suppose that potential customers arrive in accordance with a Poisson process having rate λ , and that a potential customer will enter the system only if both chairs are empty.
- Define a CTMC and find the limiting probabilities.



Balance equation: (Find by using CTMC)

$$\pi_2 \mu_2 = \pi_0 \lambda$$

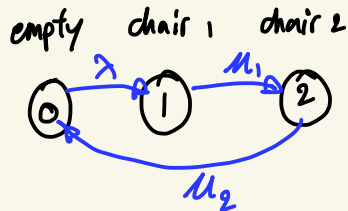
$$\pi_0 \lambda = \pi_1 \mu_1$$

$$\pi_1 \mu_1 = \pi_2 \mu_2$$

$$\pi_0 = \frac{1/\lambda}{\frac{1}{\lambda} + \frac{1}{\mu_1} + \frac{1}{\mu_2}}$$

$$\pi_1 = \frac{1/\mu_1}{\frac{1}{\lambda} + \frac{1}{\mu_1} + \frac{1}{\mu_2}}$$

$$\pi_2 = \frac{1/\mu_2}{\frac{1}{\lambda} + \frac{1}{\mu_1} + \frac{1}{\mu_2}}$$



$$Q = \begin{pmatrix} -\lambda & \lambda & 0 \\ 0 & -\mu_1 & \mu_1 \\ \mu_2 & 0 & -\mu_2 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

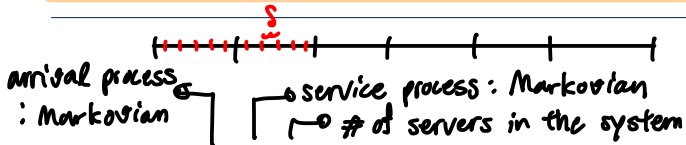
Find by using Embedded DTMC: $\pi_i^0 = \frac{1}{3} \rightarrow$ Embedded

$$\tilde{\pi}_0 = \frac{1/3}{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}}, \quad \tilde{\pi}_1 = \frac{1/3}{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}}, \quad \tilde{\pi}_2 = \frac{1/3}{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}}$$

These 2 results are the same.

$\neq$

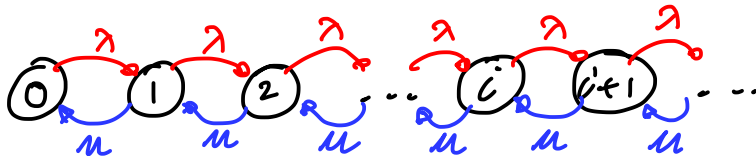
Uniformization: Discretization of CTMC to DTMC



Consider an M/M/1 queue example: Customers arrive according to a Poisson process of rate $\lambda > 0$. There is a single server and the service times are i.i.d. exponential with parameter $\mu > 0$.

Model by flipping two coins simultaneously every δ -step:

- First coin: probability $\lambda\delta$ of “arrival”
- Second coin: probability $\mu\delta$ of “departure”



Arrival δ amount of time

$$0 \rightarrow 1 - \lambda\delta + o(\delta^2)$$

$$1 \rightarrow \lambda\delta + o(\delta^2)$$

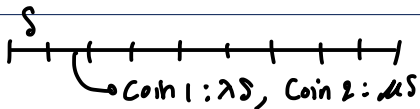
$$\geq 2 \rightarrow o(\delta^2)$$

$$0 \rightarrow 1 - \mu\delta + o(\delta^2)$$

$$1 \rightarrow \mu\delta + o(\delta^2)$$

$$\geq 2 \rightarrow o(\delta^2)$$

Uniformization: Discretization of CTMC to DTMC



Four Cases

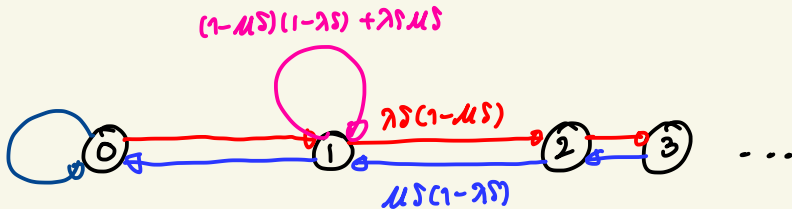
- With probability $\lambda\delta(1 - \mu\delta)$: arrival and no departure $\rightarrow$ 1 arrival, 0 completion
- With probability $(1 - \lambda\delta)\mu\delta$: departure and no arrival $\rightarrow$ 0 arrival, 1 completion
- With probability $\lambda\delta\mu\delta$: arrival and departure $\rightarrow$ 1 arrival, 1 completion
- With probability $(1 - \text{all of the above})$: no arrival and no departure $\rightarrow$ 0 arrival, 0 completion

This defines a discrete-time Markov chain that transitions every δ time units, with transition probabilities approximated by the rates above.

DTMC(S)

$$1 - \lambda\delta(1 - \mu\delta)$$

$$-\mu\delta(1 - \lambda\delta)$$



Uniformization: Discretization of CTMC to DTMC

As $\delta \rightarrow 0$, we can rewrite:

$$\lambda\delta(1 - \mu\delta) \longrightarrow \lambda\delta + o(\delta)$$

$$\mu\delta(1 - \lambda\delta) \longrightarrow \mu\delta + o(\delta)$$

$$\lambda\delta\mu\delta \longrightarrow o(\delta)$$

$$(1 - \text{all of the above}) \longrightarrow 1 - \lambda\delta - \mu\delta + o(\delta)$$

Solving the DTMC and Taking the Limit

For state 0:

$$\pi_0(\lambda\delta + o(\delta)) = \pi_1(\mu\delta + o(\delta))$$

$$\pi_0 \left(\lambda + \frac{o(\delta)}{\delta} \right) = \pi_1 \left(\mu + \frac{o(\delta)}{\delta} \right)$$

$$\lim_{\delta \rightarrow 0} \pi_0 \left(\lambda + \frac{o(\delta)}{\delta} \right) = \lim_{\delta \rightarrow 0} \pi_1 \left(\mu + \frac{o(\delta)}{\delta} \right)$$

$$\pi_0 \cdot \lambda = \pi_1 \cdot \mu \quad \Rightarrow \quad \pi_1 = \frac{\lambda}{\mu} \pi_0$$

Balance Equations for the CTMC

For state 1:

$$\pi_1(\lambda\delta + o(\delta) + \mu\delta + o(\delta)) = \pi_0(\lambda\delta + o(\delta)) + \pi_2(\mu\delta + o(\delta))$$

Taking the limit yields:

$$\pi_1(\lambda + \mu) = \pi_0\lambda + \pi_2\mu$$

Substituting $\pi_1 = \frac{\lambda}{\mu}\pi_0$:

$$\frac{\lambda}{\mu}\pi_0(\lambda + \mu) = \lambda\pi_0 + \mu\pi_2 \quad \Rightarrow \quad \pi_2 = \left(\frac{\lambda}{\mu}\right)^2 \pi_0$$

Continuing, we obtain:

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0$$

General Balance Equations for CTMCs

Balance Equations for CTMCs

For each state j :

$$\pi_j \nu_j = \sum_i \pi_i q_{ij}$$

where $q_{ij} = \nu_i \cdot p_{ij}$.

$$\pi Q = 0$$

Solving CTMC

Theorem (Summary Theorem for CTMCs)

Given an irreducible CTMC, suppose there exist π_i 's such that $\forall j$,

$$\pi_j \nu_j = \sum_i \pi_i q_{ij} \quad \text{and} \quad \sum_i \pi_i = 1.$$

Then the π_i 's are the limiting probabilities for the CTMC, and the CTMC is ergodic.

Balance $\neq$ Stationary for CTMCs

- For DTMCs, balance equations are equivalent to stationary equations.
- For a CTMC, the balance equations yield the limiting probabilities, while the stationary equations are meaningless, unless we first translate to a DTMC.

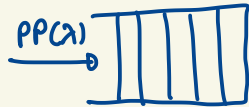
Classical Queueing Examples

Example: M/M/1 Queue

A coffee shop has **one server**. Customers arrive according to a **Poisson process** at rate λ . The service time follows an **exponential distribution** with rate μ . Find:

- (a) The limiting (stationary) distribution of the number of customers in the system
 - (b) The probability the system is empty $\rightarrow \pi_0$
 - (c) The probability there are more than 3 customers in the system
 - (d) The **throughput** (long-run departure rate) $\rightarrow \lambda$ (stable system)
 - (e) The **average queue length** (excluding the one in service)
 - (f) The **average number of customers in the system** $\hookrightarrow E[N] = E[Q] + \rho$
 - (g) The server **utilization**
- $\downarrow$
= the fraction of time the server is busy.
= $\rho = \lambda/\mu$ #
- $\downarrow \pi_n = \rho^n (1-\rho)$
- $\sum_{n=3}^{\infty} \pi_n = 1 - \pi_0 - \pi_1 - \pi_2$
- $E[Q] = \sum_{n=1}^{\infty} (n-1) \pi_n$
- $\hookrightarrow$ expected # of customers handled by the server

M/M/1

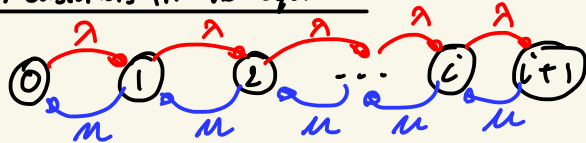


1 server



$\exp(\mu)$

of customers in the system



$$\therefore \lambda \pi_0 = \mu \pi_1 \Rightarrow \pi_1 = \frac{\lambda}{\mu} \pi_0$$

$$(\lambda + \mu) \pi_1 = \lambda \pi_0 + \mu \pi_2 \Rightarrow \pi_2 = \frac{\lambda}{\mu} \pi_1 = \left(\frac{\lambda}{\mu}\right)^2 \pi_0$$

$\vdots$

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0$$

Let $\rho = \frac{\lambda}{\mu}$;

$$\therefore \sum_{n=0}^{\infty} \pi_n = 1$$

$$\sum_{n=0}^{\infty} \rho^n \pi_0 = (1 + \rho + \rho^2 + \dots) \pi_0 = 1$$

Remark:

$$\rho < 1$$

(Convergence condition)

$$\left(\frac{1}{1-\rho}\right) \pi_0 = 1$$

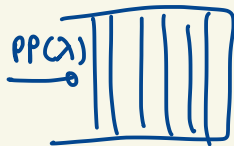
$$\therefore \pi_0 = 1 - \rho$$

M/M/c Queue

A customer service call center has c identical servers. Calls arrive according to a **Poisson process** at rate λ . Each server handles calls with service times following an **exponential distribution** with rate μ . Customers who arrive when all servers are busy join a single FIFO queue. Find $\rho < 1$

- (a) The stability condition and server utilization $\rightarrow \rho$
 - (b) The limiting (stationary) distribution of the number of customers in the system $\rightarrow \pi_n$
 - (c) The probability that an arriving customer must wait (all servers busy) $\rightarrow \sum_{n=c}^{\infty} \pi_n$
 - (d) The **throughput** (long-run departure rate) $\rightarrow \lambda$
 - (e) The **average queue length** (excluding customers in service) $\rightarrow E[Q] = \sum_{n=c+1}^{\infty} (n-c) \pi_n$
 - (f) The **average number of customers in the system**
 - (g) The probability that exactly 2 servers are busy $\rightarrow \pi_2$
- $E[N] = E[Q] + c\rho$

M/M/C



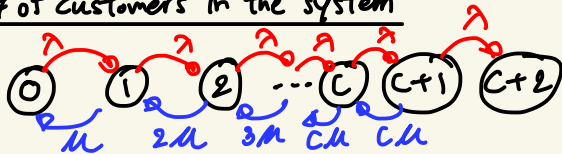
(S1) $\exp(\mu)$

(S2) $\exp(\mu)$

⋮

(Sc) $\exp(\mu)$

of customers in the system



$$\therefore \lambda \pi_0 = \mu \pi_1$$

$$(\lambda + \mu) \pi_1 = \lambda \pi_0 + 2\mu \pi_2$$

⋮

$$\text{for } 1 \leq n \leq C; \quad (\lambda + n\mu) \pi_n = \lambda \pi_{n-1} + (n+1)\mu \pi_{n+1}$$

$$\text{for } n \geq C; \quad (\lambda + C\mu) \pi_n = \lambda \pi_{n-1} + C\mu \pi_{n+1}$$

$$\therefore \text{The solution is } \pi_n = \begin{cases} \frac{(C\rho)^n}{n!} \pi_0 & ; 0 \leq n < C \\ \frac{C^C \rho^n}{C!} \pi_0 & ; n \geq C \end{cases}, \text{ where } \rho = \frac{\lambda}{C\mu} < 1.$$

$$\Downarrow \\ \sum_{n=0}^{\infty} \pi_n = 1$$